Assessing the Transferability of Text LLMs for Protein Inverse Folding via Structural Tokenization

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Abstract

This work investigates the transferability of linguistic priors from large-scale text models (Llama-3-8B) to the task of inverse protein folding. By discretizing 3D backbone structures into 1D tokens via Foldseek's 3Di alphabet and C-α coordinate-string representations, we evaluate the model's ability to design amino acid sequences. Our results indicate that while few-shot recovery performance is minimal, LoRA fine-tuning achieves improved sequence recovery, demonstrating that linguistic priors from large-scale text pre-training transfer to the protein design domain. Although overall performance remains below state-of-the-art geometric approaches, our ablation against randomly initialized model weights confirms that large-scale pretraining, and the inductive biases of large transformer architectures, provides a necessary foundation for effective adaptation to downstream tasks, without modifications to the transformer architecture.

1 Introduction

1.1 Protein Folding

Proteins are sequences of amino acids that are vital for virtually all biological processes. Protein folding is the physical process by which a one-dimensional amino acid sequence transforms into a specific three-dimensional structure, allowing it to become biologically functional. This folding is critical; even minor structural deviations can prevent a protein from interacting with its molecular targets, leading to a loss of biological activity [1]. This field saw a historic breakthrough with AlphaFold2 [2], which demonstrated that deep learning approaches could predict protein structures with near-experimental accuracy.

1.2 Inverse Protein Folding

Complementary to protein folding, Inverse Protein Folding, also known as protein sequence design, addresses the reverse mapping. Given a target 3D structural backbone, the goal is to predict an amino acid sequence that will stably fold into that specified structure. This formulation underpins many applications in enzyme and binder design, therapeutic protein development, where one starts from a desired structural scaffold and seeks amino acid sequences that realize it [3].

1.3 Related Work

Current state-of-the-art methods for inverse protein folding rely on Geometric Graph Neural Networks (GNNs) that operate directly on 3D atomic coordinates.

1.3.1 Graph Neural Network models

ProteinMPNN [3] utilizes a message-passing neural network architecture that operates on a graph where each node represents a residue and edges represent spatial proximity. To achieve rotational and translational invariance, ProteinMPNN featurizes the geometry using relative distances between C_α atoms and the relative orientations of the backbone amide groups (defined by N, C_α, C coordinates).

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PiFold [4] advances the geometric approach by introducing a physically-informed "Pi-blocks" that integrate distances, directions, and orientation angles between residues through a message-passing mechanism. It achieves superior sequence recovery rates and faster convergence than previous GNN-based architectures, including ProteinMPNN.

While these geometric models excel at capturing spatial relationships through continuous coordinate representations, they remain fundamentally specialized architectures. Their fixed inductive biases, while powerful, limit their flexibility compared to more general-purpose models. This motivates our exploration into discretizing 3D protein structures, allowing us to leverage Large Language Models (LLMs) that have demonstrated a remarkable ability to generalize across non-textual domains [5, 6] when provided with appropriate structural tokenization.

1.3.2 Protein language models

In parallel, a growing line of work on protein language models, such as ProstT5 [7] for bidirectional structure-sequence translation, train solely on amino acid sequences. This limits them to protein databases several orders of magnitude smaller than natural language corpora. Unlike text LLMs, protein language models have shown to not follow clean scaling laws [8, 9].

1.4 Motivation: Bridging the gap (from Geometry to Tokens)

A growing body of research suggests that pretrained text LLMs can generalise to complex physical domains when data is appropriately tokenized. For instance, Crystal-LLM [5] encodes crystal structures as newline-separated coordinate strings and fine-tunes LLaMA-2, demonstrating superior performance compared to the competing diffusion model CDVAE (Crystal Diffusion Variational Autoencoder) [10], without any domain-specific architectural modifications. Crucially, they find that larger pretrained LLMs better capture key crystal symmetries, suggesting that the inductive biases of LLMs are surprisingly well-suited to physical domain data.

Motivation. Rather than relying on protein-specific language models or geometry-specialized GNNs, we investigate the transferability of large-scale pre-trained text LLMs to the task of inverse protein folding. Specifically, we examine whether *(i)* language-derived priors transfer to protein design, *(ii)* performance scales across few-shot and LoRA fine-tuning regimes, and *(iii)* tokenization strategy affects downstream performance.

2 Methodology

We utilize pre-trained LLM (Llama-3-8B family) and fine-tune it on discrete string representations of protein backbone structures. The key design choice is how to discretize 3D protein structures into 1D token sequences compatible with the LLM's tokenizer. We investigate the two following strategies for this:

2.1 Structural Discretization Strategies

2.1.1 Coordinate-as-Text

Our first approach treats the raw residue C_α coordinates as standard text strings. This method leverages the observation [5] that the Llama-3 tokenizer preserves floating-point numbers as coherent tokens for coordinate representation tokenization. We focus exclusively on C_α coordinates as they provide a concise yet geometrically sufficient representation of the protein's global fold, as many state-of-the-art architectures rely on C_α traces as the primary descriptor for structural modelling [2, 3]

2.1.2 Foldseek 3Di (Latent Structural Alphabet)

We utilize the 3Di alphabet introduced by Foldseek [11]. This method employs a Vector Quantized Variational Autoencoder (VQ-VAE) to compress the local structural context of a residue, specifically the spatial orientations and distances between a residue and its nearest neighbours, into a latent space. This latent representation is then discretized into a 20-letter "3Di" structural alphabet. By using these 3Di tokens, we effectively distill the high-dimensional geometric manifold of the protein backbone into a discrete linguistic string, allowing the LLM to treat structural features as a native vocabulary.

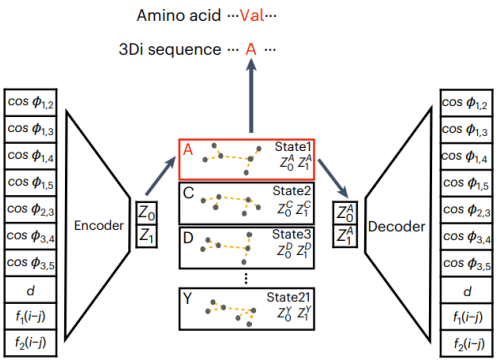

Figure 1: Foldseek learns a 20-state 3Di structural alphabet by training a variational autoencoder on local residue-residue geometry. We use its discrete latent 3Di states as a 1D structural token sequence for our LLM. Figure adapted from [11]

3 Experiments

We evaluate whether linguistic priors from large-scale pre-training transfer to the inverse protein folding task. We quantitatively compare sequence recovery performance and perplexity across tokenization strategies, model variants, and training regimes, benchmarking against domain-specific geometric state-of-the-art models ProteinMPNN [3] and PiFold [4].

3.1 Models

We evaluate two variants of Llama-3-8B [12]: the base model and its instruction-tuned counterpart, Llama-3-8B-Instruct. This model family is selected because prior work has shown that the Llama tokenizer naturally preserves the floating-point representation of 3D coordinates [5]. The instruction-tuned variant is included to investigate whether explicitly describing the structural input format in natural language improves sequence recovery.

3.2 Metrics

Sequence recovery measures the fraction of residues in the predicted sequence that match the ground-truth amino acid at each position, with higher values indicating better design quality.

Perplexity measures model confidence in the correct sequence given a structure, computed as the exponentiated average negative log-likelihood per residue. Lower perplexity indicates a sharper, more accurate predictive distribution.

3.3 Dataset

For consistency with prior work [3, 4, 7], we use CATH protein database. Each data point consists of an amino acid sequence and corresponding 3D backbone coordinates for the four main-chain atoms (N, C_α, C, O). During preprocessing, we filtered out entries in which the backbone coordinates contained continuous blocks of NaN values for any backbone atom, which would cause invalid translation for Foldseek.

3.4 Tokenization approaches (3Di vs. C_α Coordinates)

We compare the two discretization strategies introduced in section 2.1, representing protein structure as a plain-text string of C_α coordinates versus encoding it as a sequence of discrete Foldseek 3Di tokens. This comparison allows us to quantitatively assesses which tokenizer yields better downstream performance.

3.5 Learning Regimes and Data Efficiency

To assess how the model's downstream performance scales with the amount of data, we compare data efficiency comparing few-shot and LoRA fine-tuning regimes. Since each residue sequence occupies a substantial portion of the context window, limiting the number of examples that can be included in context window, the few-shot setup uses 3 examples.

LoRA finetuning. LoRA fine-tuning [13] adapts the pre-trained model by injecting trainable low-rank decomposition matrices into selected weight matrices of the transformer, while keeping the original weights frozen. We apply LoRA adapters to all projection matrices in the attention and feed-forward layers. Additionally, we include `embed_tokens` and `lm_head` as fully trainable modules to accommodate the two newly introduced special tokens `<struct>` and `<seq>`, which are initialized via mean resizing over the existing embedding distribution. We provide hyperparameters and training details in Appendix A.

3.6 Large-scale Pretraining

A central question in our work is whether performance gains stem from linguistic priors acquired during large-scale pretraining, or whether fine-tuning alone is sufficient. To isolate this, we ablate the contribution of pretraining by comparing two initialization conditions under an identical LoRA fine-tuning setup:

- **Pretrained + LoRA:** Fine-tuning Llama-3 from its pretrained weights, preserving all priors learned during large-scale language pretraining.
- **Random Init + LoRA:** Training the same architecture from a random weight initialization, effectively reducing the model to a randomly initialized transformer fine-tuned solely on our protein dataset.

The performance gap between these two conditions can be attributed to the transfer of linguistic priors from pre-training.

4 Results

4.1 Sequence Recovery across Learning Regimes and Tokenization Strategies

Table 1: Results on the CATH 4.2 dataset, reporting perplexity (↓) and sequence recovery (↑). The table compares few-shot (3 examples) and LoRA fine-tuning regimes across both 3Di and Cα tokenization strategies, for instruction-tuned and non-instruction-tuned Llama-3-8B variants, benchmarked against geometric baselines. LoRA fine-tuning with 3Di tokenization achieves the best LLM-based performance (14.08% recovery, 12.41 perplexity), demonstrating transfer of linguistic priors to the protein domain, though all LLM-based methods fall short of geometric baselines.

LLM-based methods					
Model		**Perplexity ↓**		**Recovery % ↑**	
		3Di	Cα	3Di	Cα
Llama-3-8B (non-instruct)	LoRA	12.41	17.42	14.08	5.04
Llama-3-8B-Instruct	LoRA	16.93	18.66	7.57	4.32
Llama-3-8B (non-instruct)	Few-shot (3)	18.21	19.22	4.70	1.41
Llama-3-8B-Instruct	Few-shot (3)	17.81	18.63	5.92	2.92
Model		**Perplexity ↓**		**Recovery % ↑**	
ProteinMPNN (geometric baseline)		4.61		45.96	
PiFold (geometric baseline)		4.55		51.66	

Learning regimes. Across all conditions, few-shot performance is near-random (∼2-5%), confirming that pre-trained text LLMs do not inherently interpret structural token sequences without task-specific adaptation. LoRA fine-tuning produces the most substantial gains, confirming that

gradient-based adaptation is necessary to meaningfully transfer linguistic priors for downstream tasks, confirming transfer learning. Nevertheless, all regimes fall substantially short of PiFold's 51.66%, highlighting the gap that remains between discrete-tokenization LLM approaches and purpose-built geometric architectures.

Tokenization. 3Di tokenization consistently outperforms Cα coordinate string representation across all regimes. A plausible explanation is that the 3Di tokens, being the output of a learned VQ-VAE compression mechanism, produce a more compact alphabet that is inherently easier for a language model to learn than raw floating-point coordinate strings.

Instruction-tuned vs. non-instruction-tuned. The base model consistently outperforms its instruction-tuned counterpart across both learning regimes and tokenization strategies. This suggests that instruction tuning introduces priors optimised for conversational tasks that do not transfer well to structured protein sequence prediction. Furthermore, the instruction-tuned model fails to benefit from the additional natural language descriptions of the input format.

In conclusion, non-instruction tuned Llama-3-8B with LoRA fine-tuning and 3Di tokenization, achieves 14.08% sequence recovery and a perplexity of 12.41. While this falls short of geometric baselines ProteinMPNN (45.96%) and PiFold (51.66%), it represents a meaningful improvement over the near-random few-shot baseline ($\sim$5%), demonstrating that transfer of linguistic priors to the protein design domain is achievable without any domain-specific architectural modifications.

4.2 Effect of Large-scale pre-training

The randomly initialized weights model fails to produce coherent sequences of the correct length, achieving only 4.54% recovery and a perplexity of 18.72, compared to 14.08% recovery and 12.41 perplexity for the pretrained counterpart under identical fine-tuning. This confirms that it is the combination of pre-trained linguistic priors and gradient-based adaptation that drives the transfer learning observed in our main results, in Table 1.

Table 2: Effect of weight initialization on CATH 4.2 (3Di tokenization, Llama-3-8B base). Both models use identical LoRA fine-tuning configuration. Randomly initialized weights achieve near-random sequence recovery (4.54%), compared to 14.08% for the pretrained counterpart, confirming that large-scale linguistic pretraining is a necessary foundation for effective adaptation to protein structure tasks.

Initialization	**Perplexity** $\downarrow$	**Recovery %** $\uparrow$
Pretrained + LoRA	12.41	14.08
Random Init + LoRA	18.72	4.54

5 Limitations and Future Work

The main limitation of our work is that our experiments are restricted to a single model family (Llama-3-8B). Crystal-LLM [5] achieved results competitive with state-of-the-art geometric models in crystal domain only at 70B scale. Due to computational constraints, we leave the exploration to future work whether sequence recovery in our setting similarly improves with scale, consistent with LLM scaling laws.

Beyond model size, our evaluation covers two structural tokenization strategies. Alternative discretization approaches, such as geometric byte-pair encoding (BPE)[14], remain unexplored. Hierarchical encodings are particularly interesting as they could capture structural patterns at multiple levels of granularity, from local backbone structure to global fold topology.

6 Conclusion

This work investigates whether linguistic priors from large-scale text pre-training transfer to inverse protein folding, adapting Llama-3-8B[12] via discrete structural tokenization methods without architectural modifications.

Our experiments reveal that few-shot performance is near-random, while LoRA fine-tuning produces the most significant gains, though performance lags behind geometric baselines ProteinMPNN [3] and PiFold [4]. Nevertheless, our results support the broader thesis that pretrained LLMs, paired with appropriate tokenization can complement purpose-built geometric models and be adapted to domain specific downstream tasks.

Crucially, we show that pretrained weights consistently outperform randomly initialized counterparts under identical fine-tuning conditions, highlighting the importance of large-scale pretraining, in terms of large scale data, optimization, and the inductive biases of large transformer architectures, as a necessary foundation for effective adaptation to protein structure domains.

We argue that with larger model scale and improved tokenization methods, the use of general-purpose transformer architectures for domain-specific downstream tasks represents a promising direction. Unlike purpose-built geometric models, pretrained LLMs are not constrained to a single domain and could naturally extend to multimodal and multitask settings.

GitHub repository

The companion source code for this project may be found at: https://github.com/oszkarurban/gdl-inverse-protein.

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A LoRA Training Details

We provide training details:

- Llama-3-8B: Effective batch size of 512 for 25 epochs with a cosine annealed learning rate of 3e-4. LoRA rank 16 and alpha 32.

Additionally, we experimented with rank rank r=8 and 32. We observed no substantial performance gain beyond rank 16.